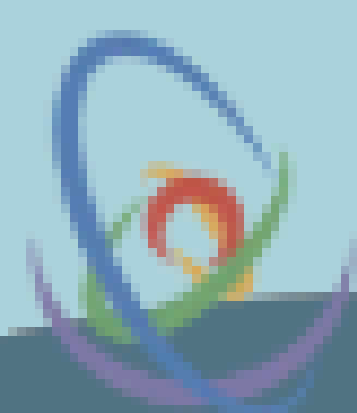




ECO-FRIENDLY SUPER CAPACITORS BY UTILIZING COCONUT FIBERS FOR HIGH-PERFORMANCE CARBON ELECTRODES

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IRC-SES

INTRODUCTION

Research Problem:

Current energy storage problems:

- Costly
- Non-sustainable materials
- Toxic

There is a growing need for:

- Sustainability
- High-performance
- Environmental friendly

Background:

Super Capacitors Offer :

- High power density
- Long lifespan
- Thermal efficiency
- Fast charging

The Project Aims :

- Fabrication of a high performance super capacitor device by using recycled materials.
- Designed to store and deliver energy efficiently showcasing the practical use of agricultural waste in green energy applications.

Objectives:

- Develop high performance super capacitor utilizing coconut fiber for cost effective, sustainable, and environmental friendly alternative.
- Optimizing parameters and investigating its electrochemical performance with H_2SO_4 as the electrolyte.
- Evaluating performance for long term super capacitor.

METHODOLOGY

1

Activated Carbon Preparation



2

Activated Carbon Preparation 2.0



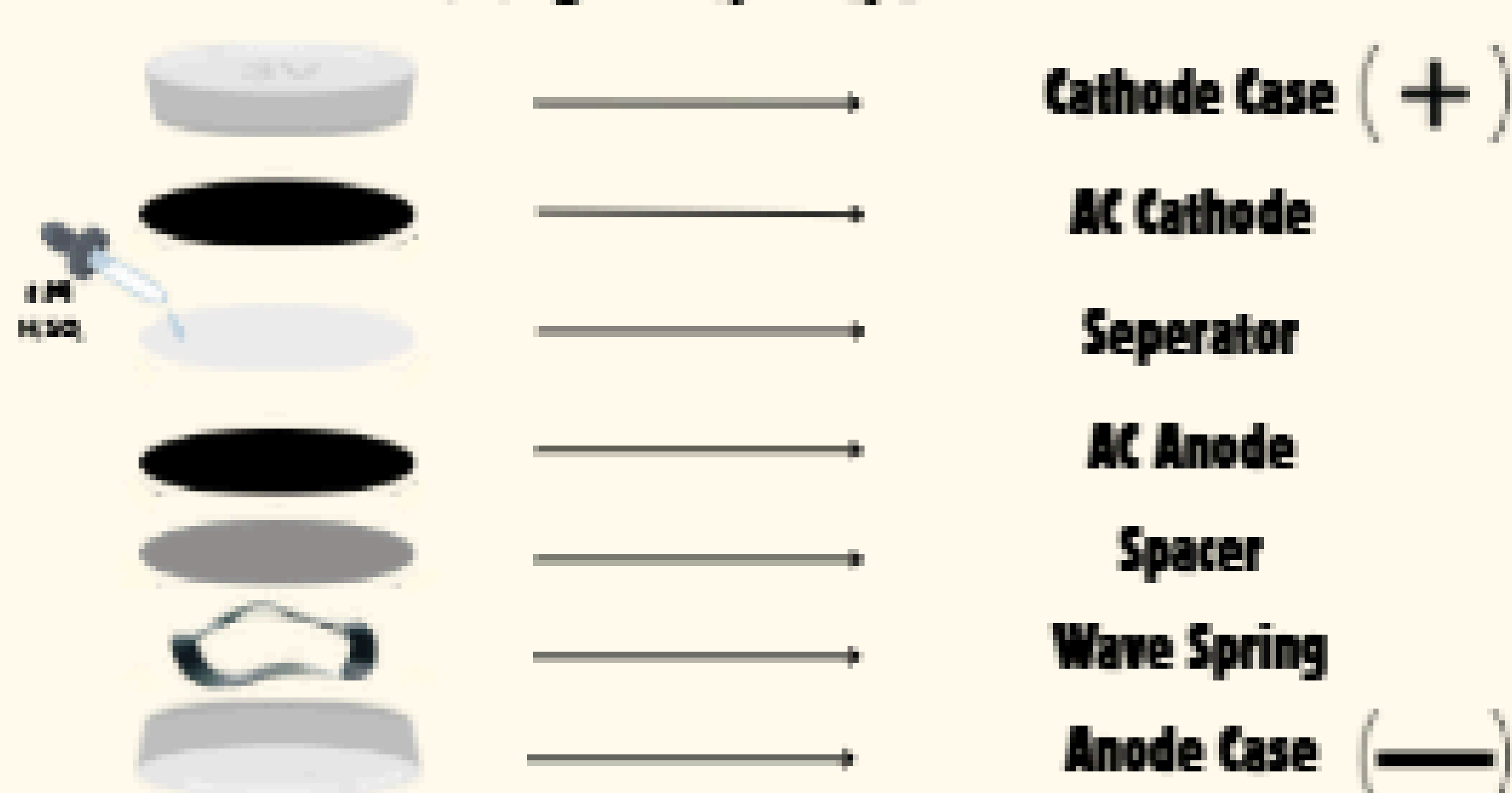
3

Fabrication of Electrodes

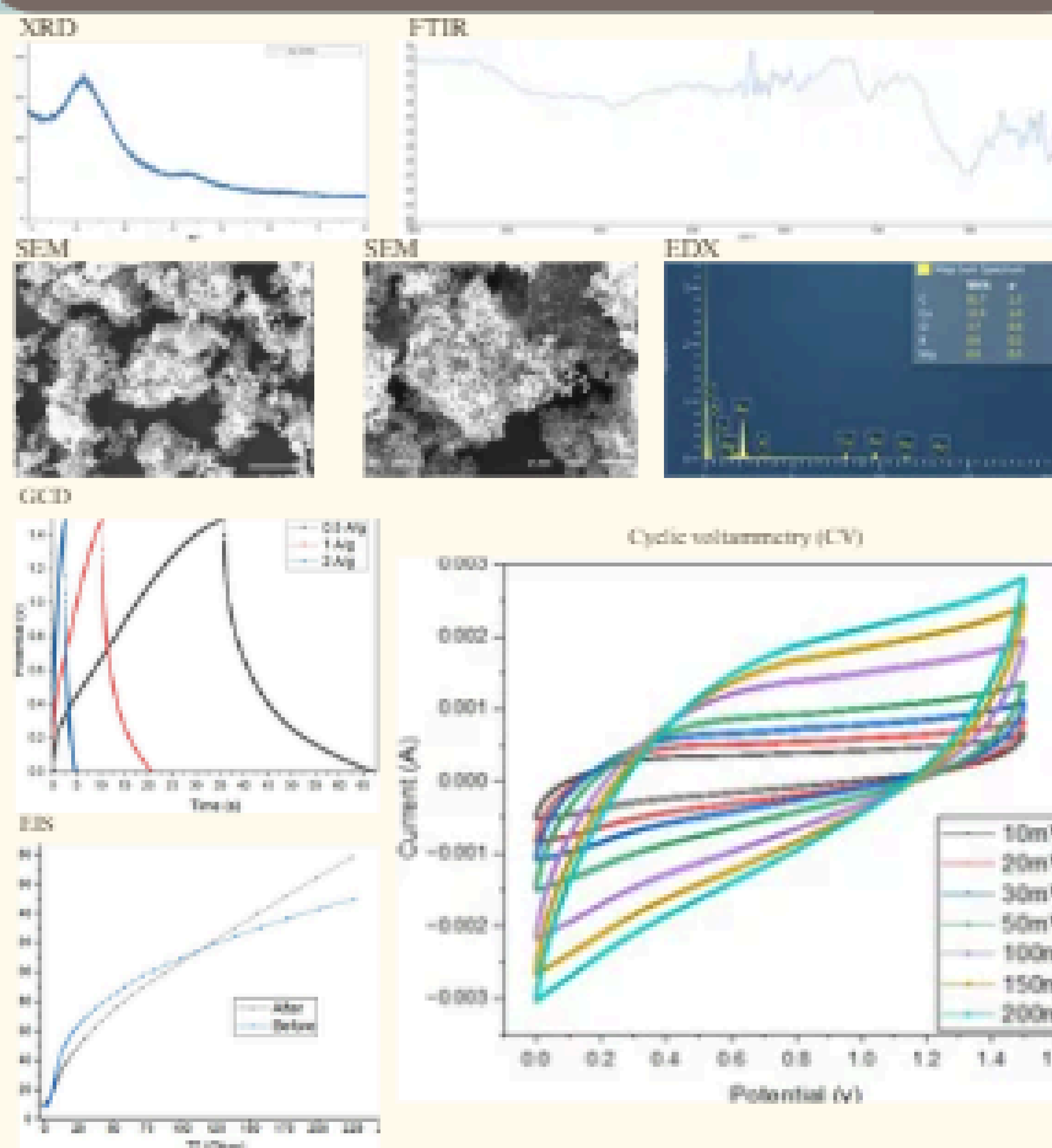


4

Building the Super Capacitor



RESULTS



CONCLUSION

Activated carbon derived from coconut fiber demonstrates strong potential as a sustainable, cost-effective electrode material for supercapacitors. When paired with sulfuric acid as the electrolyte, the system achieves high ionic conductivity and reliable electrochemical performance. Results show good capacitance, stability, and efficiency across varying acid concentrations. This approach supports eco-friendly energy storage by utilizing agricultural waste and minimizing dependence on synthetic resources, contributing to the advancement of green energy technologies.



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